

My mother always took the opportunity to teach my family important life lessons, no matter how pressing the circumstances. When I was in just the first grade, my mother signed every one of her children up for baseball or softball. She is a go-getter. She was always the team mom and greatest support for her children and team. As team mom, she had the responsibility to gather funds to pay for the team's jerseys. One day during practice, after vigorously begging each of the parents and collecting a series of checks, she left the park feeling accomplished. Everything was going to be great— the team was going to get incredible jerseys— when, after further examination, she realized that she had misplaced one of the parent's checks at the park. No worries. The only problem that could arise from this is if the parent who wrote the lost check just so happened to be the starting catcher on the Boston Red Sox when they broke the Curse of the Bambino, 3x All-Star, 2x World Series Champion, Golden Glove and Silver Spike winner, Boston Red Sox Hall of Famer Jason Varitek. And by my mom's luck, she did lose Jason Varitek's check. Every scenario came running through her head of someone using the check to drain his bank account or steal his signature and thus his identity. In order to repay the debt my mother created, I have decided to validate one of the most influential years in the history of Jason Varitek baseball, and the baseball world in general.

The Curse of the Bambino was the saving grace for much of the baseball world. Since the Red Sox team's conception, they have been incredible, dominating the MLB and winning 5 of the first 15 World Series titles. A major contributor to this success was GOAT of American Baseball, the great Babe Ruth. After a successful championship run in 1918, the owner of the Boston Red Sox decided that the payroll for Ruth was too much, and traded him to the New York Yankees, a poverty franchise, in exchange for cash. This is widely regarded as the worst trade of all time in the history of sports, ever. Do not trade the GOAT in any particular sport in their athletic prime. This trade was so horrendous, that the MLB baseball Gods decided that the Red Sox needed to be punished for their atrocities. As a result, the Boston Red Sox franchise was struck by the Curse of the Bambino, a curse that resulted in an 86 year championship drought for the franchise, only to be broken by a bloody sock and the only reverse sweep in MLB postseason history. My family were inner city Boston residents during the curse era, and had their eyes glued to the TV when the curse was lifted. They claim that the curse was real. Despite their confidence there are still many that wonder: "Is the Curse of the Bambino real?" "Is there any statistical evidence to show that during the Curse of the Bambino between 1918 and 2003 the Boston Red Sox performed significantly worse compared to the rest of the league than they did both before and after the Curse of the Bambino during both the regular and postseason?" I'm glad you asked. That is exactly what I sought out to do.

The Lahman Baseball database was crucial in finding if the Curse of the Bambino was real. It gave me the statistics, both offensive team performance using the "Team" data set, and individual postseason offensive performance using the "BattingPost" data set, that I could mutate to use the data I need through both the regular and post seasons to determine the legitimacy of the curse. By combining the two data sets, I could get a holistic picture on the impact of the Curse of the Bambino during the entire tenure of the Boston Red Sox franchise.

I decided that to provide compelling evidence about the legitimacy of the Curse of the Bambino I could measure Boston's offensive output compared to the rest of the MLB before the curse era, during the curse era, and after the curse era. To do so, I measured On-Base Plus Slugging, a metric created to show individual batting performance and general offensive

production, but on a team scale. OPS is not an individual statistic recorded in the Lahman database. To do so, I created OPS using the current formula for OPS:

$$OPS = \frac{AB * (H + BB + HBP) + TB * (AB + BB + SF + HBP)}{AB * (AB + BB + SF + HBP)}$$

When running through my code, an issue arose. Both “Hit By Pitch,” or HBP and “Sacrifice Fly,” or SF were not recorded in all of the years since the creation of the Red Sox franchise. If I was going to measure OPS over history, I wanted to ensure that the formula was consistent, and all variables could be recorded data. Because of this I opted to use the old OPS formula, before SF and HBP were recorded in 1954:

$$OPS_{\text{historic}} = \frac{AB \cdot (H + BB) + TB \cdot (AB + BB)}{AB \cdot (AB + BB)}$$

After creating this metric for both the post season and regular season and combining them into one uniform tidy set of data, I decided to compare Boston’s Historic OPS to those of other MLB teams before, after, and during the era of the Curse of the Bambino. I wanted to compare OPS against other MLB teams, and not just measure change over time because offense in baseball changes over time. By comparing the OPS within each era to the rest of the MLB, we mitigate the impact of offensive production change as time goes on. Using all of this context, I created an ANOVA table to measure the impact of season type (regular versus post season), era (before the Curse of the Bambino, during the curse, and after the curse), and team group (Boston Red Sox versus the rest of the league). This was the result:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
era	2	1.650	0.8250	229.184	< 2e-16	***
team_group	1	0.094	0.0937	26.025	3.57e-07	***
season_type	1	1.661	1.6609	461.388	< 2e-16	***
era:team_group	2	0.027	0.0133	3.693	0.025008	*
era:season_type	2	0.052	0.0261	7.257	0.000717	***
team_group:season_type	1	0.002	0.0019	0.521	0.470318	
era:team_group:season_type	2	0.011	0.0053	1.467	0.230658	
Residuals	3192	11.490	0.0036			

Signif. codes:	0	‘***’	0.001	‘**’	0.01	‘*’
				0.05	‘.’	0.1
					‘ ’	1

Incredible. From this table we can learn many things: Team OPS is significantly different depending on era - This is expected because offensive production changes as time goes on. More explosiveness, more hits, higher overall OPS; Boston’s OPS is significantly different than the rest of the league - This is also expected. There are a lot of poverty franchises in the MLB.

Boston is a huge market, so their team OPS is significantly better than the rest of the league average; and Team OPS significantly differs in the playoffs for the entire league average - This once again is expected. Because bottom feeders don't make the post season, league average OPS is going to be higher in the postseason because teams with higher OPS are more likely to make the playoffs because offense wins baseball games.

There is however data that is more pertinent to my cause of determining the legitimacy of the Curse of the Bambino. Line number 4:

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era:team_group          2  0.027  0.0133  3.693 0.025008 *
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This line shows that the Red Sox did significantly differently than the rest of the MLB depending on the era over combined regular and postseason. Look at that star. Because the p-value of .025008 is lower than .05, we reject the null hypothesis, meaning there is a SIGNIFICANT DIFFERENCE in Red Sox team historic OPS compared to the rest of the MLB depending on era. After further analysis, or just understanding the curse, we can say that the 1919–2003 Red Sox did not perform relative to MLB the same way they did before the curse and after the curse in regards to team historic OPS.

Although this data cannot prove that there was some supernatural element to the curse, it can prove that the offensive production of the Red Sox was worse during the Curse of the Bambino. For some, they can point to psychological choke factors, roster construction, or luck to account for this difference, but for lifelong Boston Red Sox fans, we can use this as evidence to prove that the Curse of the Bambino was real. Thank goodness it is over.

Additional Notes:

- Jason Varitek just wrote my mom another check. Nothing bad happened to his identity.
- Many of the assumptions made after the ANOVA table are dependent on a graphic that I will attach separately. It's just a yucky graphic, so it won't be concluded in my write up.
- This project was fun! Sorry for the long report